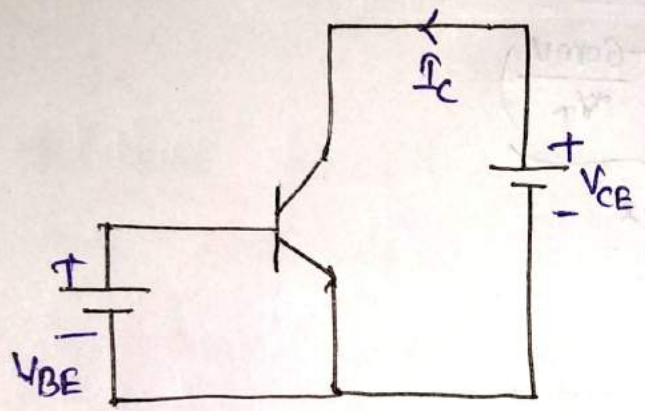
Lecture – 14

• Basic Properties of Bipolar Transistors:-



$$I_C = I_S\left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1\right]$$

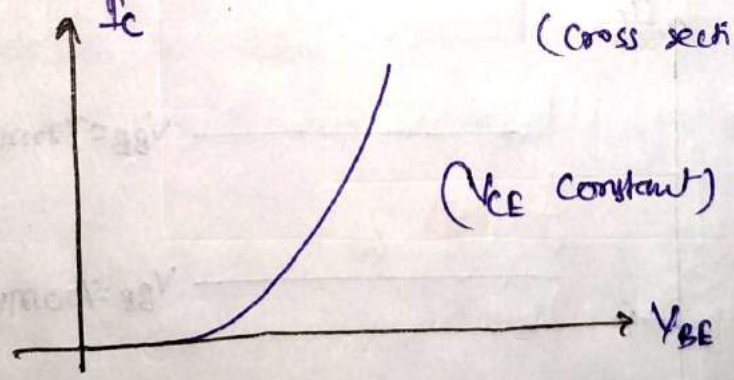
Where I_S is given by

$$\boxed{I_S = \frac{A_E \cdot q \cdot n_i^2 \cdot D_n}{W_B \cdot N_D}}$$

Emitter Area (cross sectional)

Doping in Base

Base Width



* Since we have two voltages so we have to define it when we plot waveforms.

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Que: How much does I_C change if V_{BE} increases by 60 mV?

Current I_C changes by factor of 10 if V_{BE} increases by 60 mV like diode.

* In most cases,

$$I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

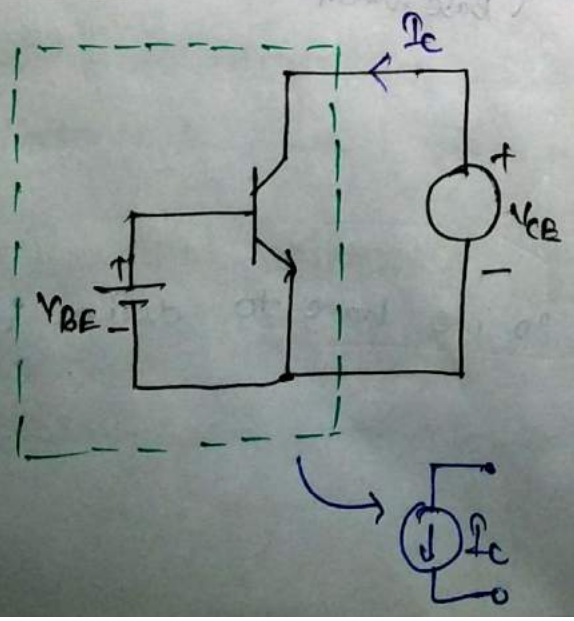
$$\Rightarrow \boxed{V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)}$$

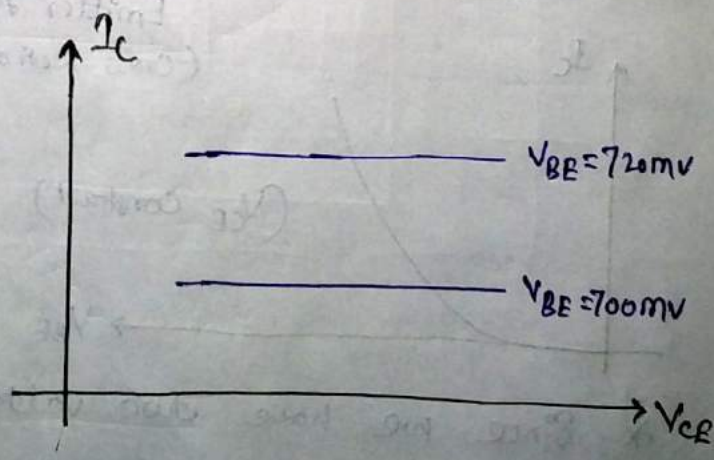
Example: We double the area of a transistor and decrease its base emitter voltage by 60mV. What is the change in the collector current?

$$I_S \exp\left(\frac{V_{BE}}{V_T}\right) \longrightarrow 2I_S \exp\left(\frac{V_{BE}}{V_T}\right) \underbrace{\exp\left(\frac{-60mV}{V_T}\right)}_{0.1}$$

$$= 0.2 I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

• I_C v/s V_{CE}





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* The collector current does not effected much by V_{CE} so we can say that it will be constant.

* The circuit in box works as current source because it provides a constant current I_C. However, when we change V_{BE} from 700mV to 710mV then the current increases but its constant because V_{BE} is not varying. Therefore we may conclude that the transistor is Voltage dependent current source because in this collector current I_C is dependent on V_{BE} voltage level.

* Since Transistor acts as a Voltage dependent current source so it can be used as an amplifier.

Terminal Currents:-

$$I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

* Since at the Base-Emitter junction, lots of electrons go to the base and few holes go to emitter which are provided by Base that means there must be a current due to holes. Sometimes some holes recombine with electrons at base so collector current reduces and we may conclude that collector current is affected by base current.

$$I_B \propto I_C \Rightarrow \boxed{I_C = \beta I_B}$$

β : 50 – 200

β → Current gain

Therefore,

$$\boxed{I_B \approx \frac{I_S}{\beta} \exp\left(\frac{V_{BE}}{V_T}\right)}$$

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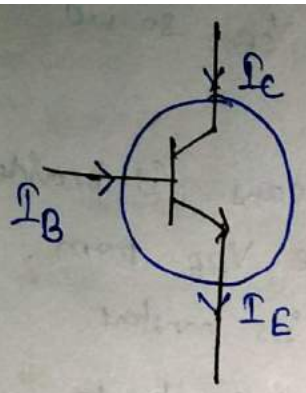
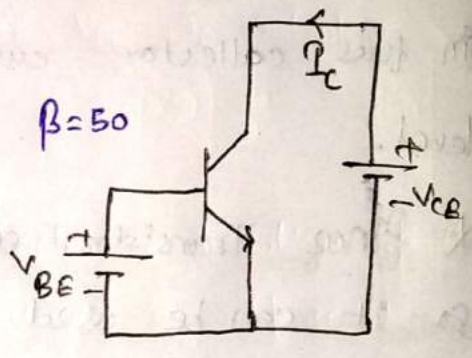


So by KCL,

$$I_E = I_C + I_B = \beta I_B + I_B = (\beta+1) I_B$$

$$I_E \simeq \frac{\beta+1}{\beta} I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

Example: A Bipolar transistor has a collector current of 1 mA and $I_S = 10^{-16}$ Amp. How much is V_{BE}?



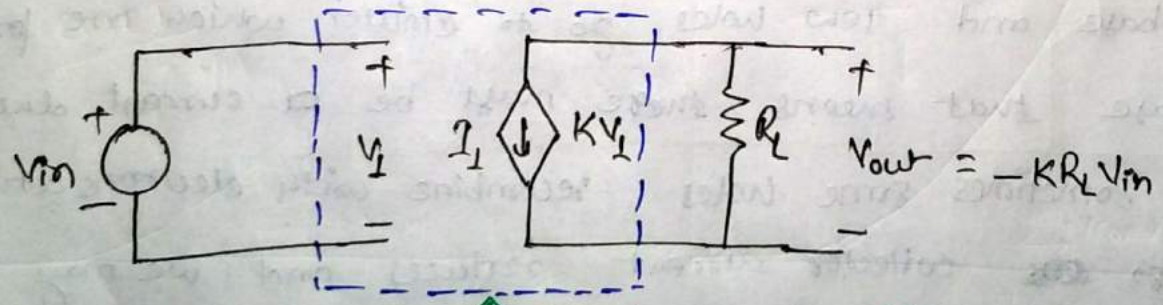
$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$$

$$= 26\times10^{-3} \ln\left(\frac{10^{-3}}{10^{-16}}\right)$$

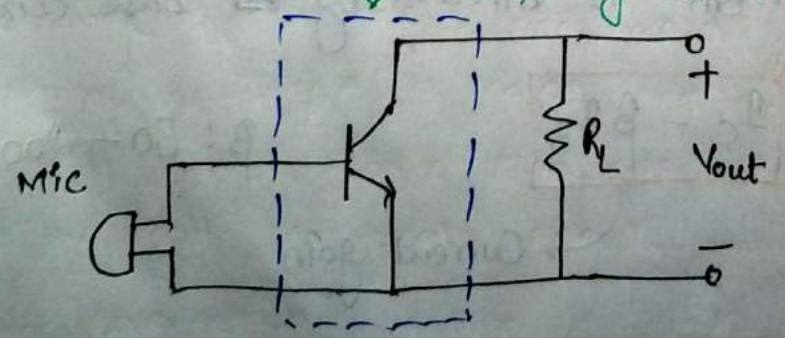
$V_{BE} =$ 0.778 V

$$I_B = \frac{I_C}{\beta} = \frac{1}{50} = 20\mu A$$

- Let's Build an amplifier:-



⇕ Corresponding ckt of it



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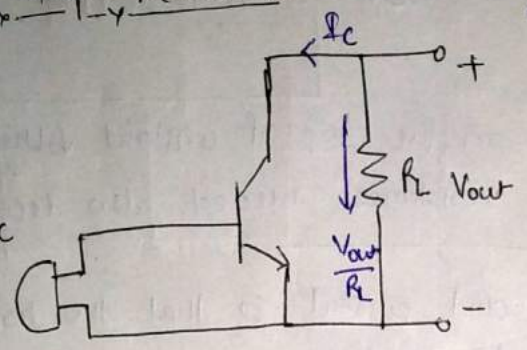
Lecture – 15

Simple Amp. Revisited:-



Say, $I_s = 5\times10^{-16}$ Amp

$$I_c = -\frac{V_{out}}{R_L}$$

$$\Rightarrow V_{out} = -R_L I_c$$

$$= -R_L I_s\left[\exp\left(\frac{V_{BE}}{V_T}\right)-1\right]$$

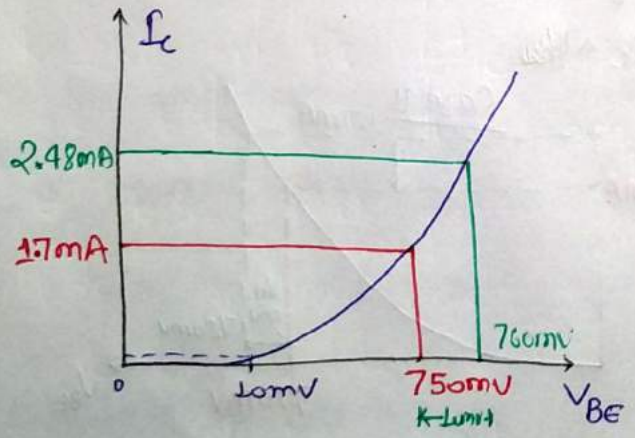
This equation tells that when the V_{BE} changes then V_{out} changes. When the V_{BE} due to mic becomes 10mV from 0V then by this equation V_{out} becomes approx 100mV i.e. amplified version of V_{BE}.

$$V_{out} = -R_L \times \underbrace{5\times10^{-16}\left(\exp\left(\frac{10}{26}\right)-1\right)}_{2.39\times10^{-16}\,A}$$

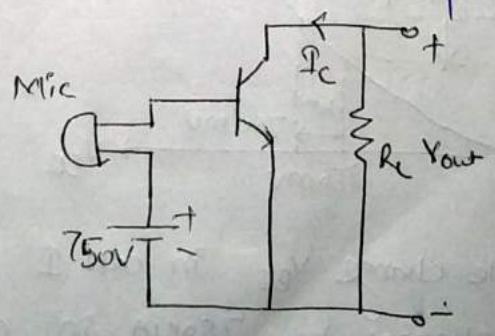
for $V_{out} = -100\,mV$

$$\Rightarrow R_L = 4.25\times10^{16}\,\Omega$$

Since R_L is so large which is not possible due to not availability of it in market.



* As shown in figure, the current I_c is tiny at 10mV. So for fixing this we just connect a battery in series with microphone.



* When we are speaking at mic so mic produces 0 Volts so we have 750 mV only and collector current I_c 1.7 mA.
* When we speak at the mic then it produces 10 mV then voltage V_{BE} becomes 760 mV then collector current becomes 2.48 mA. So when the voltage is increased by 10 mV then I_c ↑ by 780 μA.

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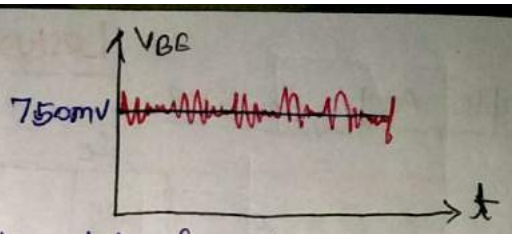
for getting 100mV at output

$$R_L = \frac{100mV}{780\mu A} = 128\Omega$$



* If we superimpose the microphone on the top of constant voltage then we start with a good current and resistance needed also decreased.

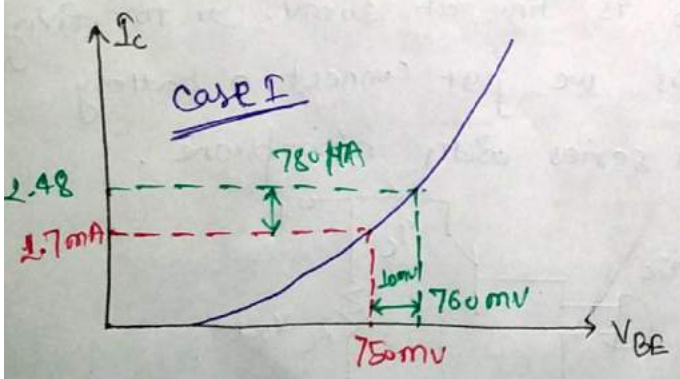
Biasing:- Provide proper voltage and current so that the transistor can amplify. (In the absence of signals)

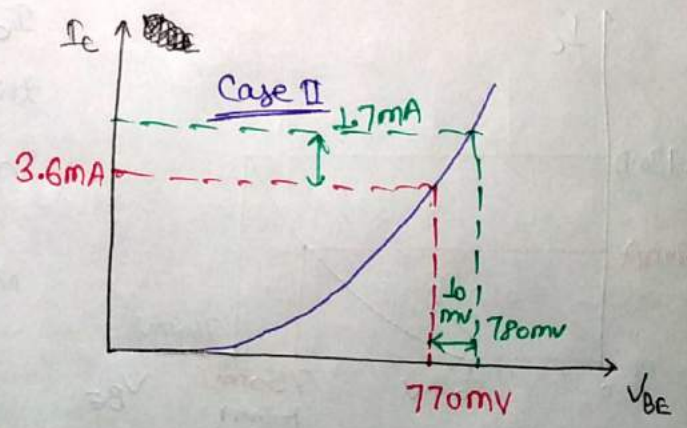
Operating point:- The bias value chosen for V_{BE}, I_C and other values, e.g. ($V_{BE} = 750mV$, $I_C = 1.7mA$).

Observation:- ① A Bipolar transistor can acts as a voltage dependent current source ⇒ Its collector current changes in response to its base emitter voltage.

② The operating point determines how the transistor responds.

③ Which operating point is "better"?





→ If we change V_{BE} in case I and case II by 10mV then collector current changes by 780μA in case I and 1.7mA in case II. So case II has better response than case I. Then amplifier amplifies the signal to greater extent.

→ If the base-emitter voltage is greater then the current I_C becomes more sensitive in the response of V_{BE}.

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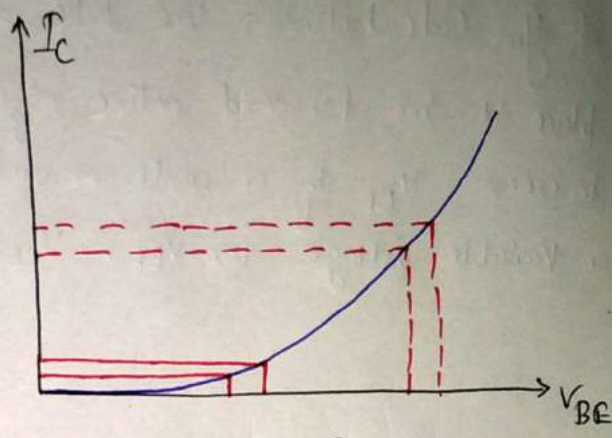
Concept of Transconductance:-



$$g_m = \frac{dI_C}{dV_{BE}} = \text{(Transconductance)}$$

dV_{BE} ← small change in V_{BE}

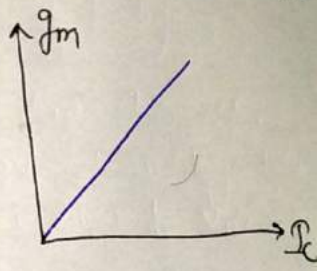
Approximate result

For case I,

$$g_{m_1} \cong \frac{780\,\mu A}{10\,mV} \cong 78\,mA/V \text{ or } mS$$

For case II,

$$g_{m_2} \cong \frac{1.7\,mA}{10\,mV} \cong 0.17\,A/V \text{ or } S$$



since $I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)$ therefore,

$$g_m = \frac{d}{dV_{BE}}\left[I_S \exp\left(\frac{V_{BE}}{V_T}\right)\right] = \frac{I_S}{V_T}\exp\left(\frac{V_{BE}}{V_T}\right) = \frac{I_C}{V_T}$$

I_C

$$\boxed{g_m = \frac{I_C}{V_T}}$$

for example, $I_C = 1\,mA \Rightarrow g_m = \frac{1\,mA}{25\,mV} = \frac{1}{25\,\Omega} = 0.04\,S$

⇒ With no bias, $I_C = 0 \Rightarrow g_m = 0 \Rightarrow$ no amplification

⇒ For amplification need a certain $g_m \Rightarrow$ certain $I_C \Rightarrow$ certain V_{BE}

For accurate result of case I & case II,

$$g_{m_1} = \frac{I_{C_1}}{V_T} = 0.065\,S \qquad g_{m_2} = \frac{I_{C_2}}{V_T} = 0.138\,S$$

⇒ Trade off between g_m and power consumption ⇒ For getting higher g_m we increase current so power consumption also increases, since we have not infinite power so we have to consider power consumption also.

⇒ A "small" change in $V_{BE} \cong$ a small fraction of V_T.

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